

Mingyang Jiang

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Education

Vanderbilt University

Nashville, TN

Bachelor of Science in Computer Science

Expected Dec 2026

- GPA: 3.932/4.0
- **Relevant Coursework:**
 - *Machine Learning & AI:* Applied Machine Learning, Formal Methods: Safe and Trustworthy AI, Natural Language Processing for Healthcare
 - *Systems & Software:* Intermediate Software Design, Computer Architecture
 - *Mathematics & Theory:* Linear Algebra, Algorithms, Probability and Statistics for Engineering
 - *Applications:* Network Analysis in Healthcare

Papers and Posters

- Jiang, M., Ni, C., Liu, W., & Yin, Z.. **Characterizing Treatment-Context Medication Evidence Across Clinic Notes and Structured EHR Medication History.** *Submitted to the IEEE International Conference on Bioinformatics and Biomedicine (BIBM) 2026.* [arXiv:2608.01570](#)
- Jiang, M., Ondov, B., Chang, C.-H., & Xu, H.. **Source-Grounded and Auditable Contract-First Generation of Clinical Transition Notes from Electronic Health Records.** *Manuscript in preparation; intended submission to npj Digital Medicine.*
- Jiang, M., Li, Y., Moyer, D., Xu, H., & Chang, C.. **Dataset-Dependent Effects of Cross-Depth Aggregation and Soft-Routed Experts in EEG Foundation Model Fine-Tuning.** *Submitted to the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2027.* [arXiv:2609.17886](#)
- Jiang, M., Li, Y., Moyer, D., Ma, F., Xu, H., & Chang, C.. **Testing Backbone-Informed Interaction Adapters for EEG Foundation Models: A Capacity-Controlled Study.** *Submitted to Transactions on Machine Learning Research (TMLR).*
- Jiang, M., Ni, C., & Yin, Z.. **Note-to-Structured EHR Concordance: Pipeline and Drug Normalization.** *Accepted poster at AMIA 2026.*
- Peng, X., Qian, L., Wang, Y., Xiang, R., He, Y., Ren, Y., Jiang, M., Zhang, V. J., Guo, Y., Zhao, J., & others. **MultiFinBen: Benchmarking Large Language Models for Multilingual and Multimodal Financial Application.** *Accepted to ACL 2026 (Main Conference).* [arXiv:2506.14028](#)
- Ma, F., Giuffrè, M., Wright, D., McCann, K., Iscoe, M., Qian, L., Jiang, M., Ng, C. W., Hong, N., He, H., Shyr, C., Chen, Q., Schwamm, L., Ohno-Machado, L., & Xu, H.. **A Safety-Oriented Hypothetico-Deductive Framework for AI-Assisted Differential Diagnosis.** *Major revision invited at Nature Communications.* [arXiv:2607.08038](#)
- Ma, F., An, Q., Chen, P., Qian, L., Lan, X., Jiang, M., Gu, Z., Papademetris, X., & Xu, H.. **TRACE: Temporal Routing with Autoregressive Cross-channel Experts for EEG Representation Learning.** *Resubmitted to ICLR 2027.* [arXiv:2605.11380](#)
- Peng, X., Xiang, R., Zhang, F., Song, M., Jiang, M., Wang, Y., Qian, L., Hara, T., Guo, Y., Huang, J., Tsujii, J., & Ananiadou, S.. **Ebisu: Benchmarking Large Language Models in Japanese Finance.** *Resubmitted to ACL ARR (August 2026); under review.* [arXiv:2602.01479](#)
- Peng, X., Xie, Z., Cao, Y., Li, H., Qian, L., Wang, Y., Zhang, V. J., He, H., Ai, X., Ma, L., Xiang, R., He, Y., Wang, S., Guo, Y., Jiang, M., & others. **HERCULEAN: An Agentic Benchmark for Financial Intelligence.** *Resubmitted to ACL ARR (August 2026); under review.* [arXiv:2605.14355](#)

Research Experience

Research Intern

June 2025 – Present

Clinic NLP Lab, Yale School of Medicine (BIDS), New Haven, CT

- **Project 1: TRACE – EEG Foundation Model**

- Contributed to TRACE, an autoregressive EEG foundation model with temporal routing and cross-channel expert sharing for representation learning across heterogeneous montages and recording settings
- Collected, cleaned, and preprocessed multi-source EEG data for large-scale pretraining and downstream evaluation; implemented baseline EEG foundation models for comparison
- TRACE achieved leading results on several downstream EEG benchmarks and was resubmitted to ICLR 2027

- **Project 2: Source-Grounded Clinical Transition Note Generation**

- Developing a contract-first pipeline that transforms credentialed-access, de-identified EHR discharge records into auditable clinical transition notes using LLM-assisted course generation and deterministic rendering of high-risk discharge information
- Built provenance-preserving extraction, medication/regimen atomization, patient-disjoint evaluation, and refusal rules for incomplete or contradictory transitions
- Designed held-out evaluation for factual faithfulness, clinically material omissions, representation support, and downstream utility; conducting lexical and semantic copying/memorization characterization
- Preparing blinded clinician evaluation of source-faithfulness, safety, and usability
- Generated outputs remain restricted source-grounded transformations and are not presented as privacy-preserving or publicly releasable

- *Keywords:* Clinical NLP, EEG signal processing, foundation models, EHR data curation, LLM evaluation, clinical text generation, provenance, privacy auditing, synthetic health data

VUSE Research Fellow

February 2026 – Present

Neurdy Lab, Vanderbilt University, Nashville, TN

- Conducting EEG foundation model research under PI Dr. Catie Chang, with a focus on efficient downstream adaptation across heterogeneous transfer tasks
- Developed and evaluated depth-aware selective adaptation methods for pretrained EEG backbones using lightweight specialist modules and task-aware routing
- Contributed to the paper *Dataset-Dependent Effects of Cross-Depth Aggregation and Soft-Routed Experts in EEG Foundation Model Fine-Tuning*, submitted to the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2027
- Completed experiments for *Testing Backbone-Informed Interaction Adapters for EEG Foundation Models: A Capacity-Controlled Study*, submitted to TMLR
- *Keywords:* EEG foundation models, transfer learning, parameter-efficient adaptation, neuroAI

Research Assistant

February 2026 – Present

SPHERE Lab, Vanderbilt University Medical Center, Nashville, TN

- Studying clinic-note and structured EHR medication concordance under PI Dr. Zhijun Yin through note-grounded normalization and cross-source comparison
- Developed a deterministic normalization and evaluation framework combining LLM-assisted candidate generation with semantic and temporal mismatch analysis for treatment-context medication evidence

- Wrote the full manuscript *Characterizing Treatment-Context Medication Evidence Across Clinic Notes and Structured EHR Medication History*, submitted to the IEEE International Conference on Bioinformatics and Biomedicine (BIBM) 2026
- Produced an AMIA 2026 accepted poster presenting the project's methodology and findings

Research Contributor

April 2025 – June 2026

The Fin AI

- **Project 1: MultiFinBen - Multilingual and Multimodal Financial Benchmark**
 - Contributed to MultiFinBen, an expert-annotated multilingual and multimodal benchmark for evaluating financial LLMs across five languages and multiple input modalities
 - Processed financial reports and bilingual financial news data to support cross-lingual QA task construction and benchmark evaluation
 - Evaluated multiple LLMs on PolyFiQA subsets and built comparative analyses of multilingual reasoning performance
- **Project 2: Ebisu - Japanese Financial Language Understanding Benchmark**
 - Contributed to Ebisu, a benchmark for Japanese financial language understanding focused on linguistically and culturally grounded phenomena in corporate communication
 - Collected Japanese corporate disclosure data, including IPO materials and financial reports, to support dataset construction and annotation
 - Supported evaluation and analysis showing persistent model weaknesses on pragmatic inference and term extraction tasks
- *Keywords:* Financial NLP, LLM benchmarking, multilingual data processing, financial document parsing

Teaching Experience

Invited Speaker

Summer 2026

Vanderbilt AI Summer School 2026, Vanderbilt University, Nashville, TN

- Delivered a 15-minute talk on *From Attention to Brain Foundation Models: Transformers, Transfer Learning, and EEG Time-Series Intelligence*

Teaching Assistant

Fall 2025

CS 2281: Computer Architecture, Vanderbilt University, Nashville, TN

- Supervised laboratory sessions, providing technical guidance and troubleshooting support to students on computer architecture concepts and hardware implementation
- Evaluated student performance through in-lab assessments and graded laboratory reports, ensuring consistent feedback on assignments covering processor design, memory systems, and instruction set architecture
- Coordinated make-up sessions for students who missed scheduled laboratory periods, maintaining course continuity and ensuring equitable access to hands-on learning experiences
- Collaborated with course instructor Dr. Shervin Hajiamini to align laboratory activities with course objectives and support student learning outcomes

Technical Skills

- **Programming Languages:** Python, Java, C++
- **Machine Learning & Data Science:** Pandas, NumPy, Scikit-Learn, Matplotlib, Keras, PyTorch
- **Research Areas:** Medical NLP, EEG, Foundation Models, Financial NLP, Multimodal Learning